Bridging the Higher Education–Workforce Divide: A Large-Scale Empirical Analysis of Academic–Industry Skills Misalignment Using Natural Language Processing and Machine Learning

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***Abstract*—The accelerating divergence between academic curricula and contemporary industrial skill demands poses a fundamental structural challenge to higher education and modern labor economies. While anecdotal and survey-based reports frequently cite severe graduate skill shortages, large-scale empirical quantification that bridges macro-level labor demand with micro-level employability outcomes remains scarce. This paper presents a comprehensive, data-driven investigation of academic–industry skills misalignment utilizing 15 multi-source corpora comprising 3,176,266 records spanning LinkedIn, HuggingFace, and public benchmark repositories whose observation windows cover 2020 through 2026. Grounding our empirical inquiry in Human Capital Theory, Job Market Signaling, and Constructive Alignment, we develop a multi-paradigm analytical framework comprising: (1) natural language processing skill extraction across an eight-category taxonomy; (2) semantic embedding alignment utilizing pre-trained Sentence-BERT (`all-MiniLM-L6-v2`) across four cosine similarity thresholds ($\tau \in \{0.4, 0.5, 0.6, 0.7\}$) with 1,000-fold bootstrap confidence intervals; (3) inferential econometric statistics (Welch's t-tests, Cohen's d, Cramér's V, and Variance Inflation Factor diagnostics); and (4) 5-fold stratified cross-validated machine learning classifiers complemented by diagnostic target leakage audits and independent external validation. Our findings demonstrate a critical duality: while surface lexical string matching exhibits a severe nominal Skill Gap Index (SGI) of 52.0%, dense semantic vector alignment reveals an estimated semantic skill gap of 32.0% at baseline threshold $\tau = 0.5$ (95% bootstrap CI: [23.0%, 41.0%]), which escalates to 46.0% (95% CI: [37.0%, 56.0%]) under stringent matching criteria ($\tau = 0.7$). Disciplinary deficits concentrate acutely in emerging technological domains, notably Data Science (18 percentage-point gap) and Cloud Infrastructure (15 percentage-point gap). In predictive employability modeling, standardized logistic regression identifies Communication Skills (OR $= 6.47$, $p < 0.001$), Academic CGPA (OR $= 6.20$, $p < 0.001$), and Practical Project Experience (OR $= 3.18$, $p < 0.001$) as the primary determinants of graduate placement (CV Accuracy $= 90.11\%$, ROC-AUC $= 0.9406$). Diagnostic audits reveal that near-perfect tree-ensemble performance ($\geq 99.9\%$) stems from semi-deterministic benchmark decision boundaries, with independent external evaluation on an out-of-distribution 50,000-student cohort demonstrating performance attenuation (ROC-AUC $= 0.7686$) under domain shift. We synthesize our empirical findings into actionable institutional recommendations for universities, industry recruiters, and accreditation bodies, while rigorously documenting threats to validity, curriculum proxy constraints, and data provenance.**

***Index Terms*—Skills gap, Higher education, Employability, Natural Language Processing, Sentence-BERT, Machine learning, Curriculum alignment, Human capital theory, Data provenance.**

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I. Introduction

THE rapid acceleration of Industry 4.0, distributed cloud architectures, and generative artificial intelligence has catalyzed an unprecedented structural transformation across global labor markets. As enterprise organizations transition toward data-intensive, automated workflows, the composition of skills demanded by employers is shifting at a velocity that vastly outpaces traditional academic curriculum revision lifecycles [1], [2]. Consequently, higher education institutions (HEIs) confront what educational researchers and labor economists term the *skills gap*—the systemic disparity between the competencies imparted through university degree programs and the practical, technology-driven proficiencies demanded by contemporary employers [3], [4].

According to reports from the World Economic Forum [5], over 40% of core worker skills are expected to change within five-year windows, with severe shortages emerging in artificial intelligence, distributed computing, and data engineering. Concurrently, employers frequently express dissatisfaction with graduate preparedness, highlighting deficits not only in applied technological domains but also in essential transferable soft skills, including professional communication and collaborative problem solving [6]–[8].

Despite the widespread recognition of this phenomenon, the existing scholarly literature exhibits critical method-

ological limitations:

(1) **Scale and Geographic Narrowness:** Prior investigations predominantly rely on localized institutional surveys, small employer cohorts ($N < 500$), or single-nation case studies [9], [10], failing to capture broad macroeconomic skill shifts.
(2) **Methodological Fragmentation:** Studies either employ qualitative thematic coding [11] or purely predictive machine learning without semantic text mining, neglecting the distinction between lexical terminology mismatch and genuine conceptual skill absence [12], [13].
(3) **Temporal Invalidation:** Many published baseline studies pre-date the post-pandemic digital migration and the widespread commercial integration of generative AI (2022–2026) [14], [15].
(4) **Lack of Theoretical Integration:** Empirical machine learning studies often operate in a theoretical vacuum, omitting foundational economic and educational models that explain *why* curriculum inertia persists [16], [17].

To bridge these gaps, this study presents a multi-dataset empirical framework analyzing 15 datasets comprising over 3.18 million records from LinkedIn, HuggingFace, and benchmark repositories. We formulate five core research questions:

- **RQ1 (Market Topography & Evolution):** What are the most in-demand competencies in the contemporary labor market, and how have core technical skills evolved from 2020 through 2026?
- **RQ2 (Curriculum Alignment Duality):** To what extent do academic curriculum benchmarks align with market demands when evaluated through surface lexical matching versus dense semantic vector embeddings?
- **RQ3 (Category-Level Deficits):** What is the magnitude of the skill deficit across distinct disciplinary domains (Programming, Data Science, Cloud Infrastructure, Soft Skills, Cybersecurity)?
- **RQ4 (Employability Determinants):** Which academic and co-curricular factors best predict graduate hiring outcomes, and how do linear probabilistic models compare against non-linear tree ensembles under rigorous validation?
- **RQ5 (Geographic Concentration & Cross-Cohort Contexts):** How is labor market demand geographically distributed across international postings, and how do employment transition outcomes vary across distinct student and graduate cohorts?

Contributions: This work advances the literature through five primary contributions:

1) **Massive-Scale Empirical Corpus:** We aggregate and harmonize 3,176,266 labor market and educational records across 15 datasets, establishing one of the largest cross-sectional labor-curriculum corpora in the empirical literature.
2) **Dual Lexical-Semantic NLP Pipeline:** We deploy Sentence-BERT (`all-MiniLM-L6-v2`) embedding models to demonstrate that surface lexical mismatch (52.0% gap) substantially overstates the estimated semantic skill gap (32.0% at baseline threshold $\tau = 0.5$).
3) **Bootstrap Sensitivity Analysis:** We provide empirical bootstrap confidence intervals across four cosine similarity thresholds ($\tau \in \{0.4, 0.5, 0.6, 0.7\}$), ensuring mathematical transparency.
4) **Rigorous ML Validation & Diagnostic Auditing:** We execute 5-fold stratified cross-validation across five distinct algorithmic paradigms, supported by exact Variance Inflation Factor calculations (VIF ≤ 1.001), odds ratio estimation, deterministic decision-boundary leakage diagnostics, and independent external validation on a 50,000-student cohort.
5) **Theory-Driven Institutional Framework:** Grounding our analysis in Human Capital Theory, Signaling Theory, and Constructive Alignment, we deliver concrete policy recommendations for universities, recruiters, and accreditation bodies, accompanied by transparent reporting of dataset provenance and validity threats.

The remainder of this article is organized as follows: Section II synthesizes related work and establishes the theoretical framework. Section III delineates the research design, dataset provenance, NLP architecture, and predictive modeling protocols. Section IV presents empirical findings addressing RQ1–RQ5. Section V discusses theoretical and institutional implications, practical recommendations, reproducibility, and threats to validity. Section VI concludes the study.

II. Related Work and Theoretical Foundations

A. Conceptualizations of the Skills Gap

The concept of a skills gap encompasses multi-faceted dimensions across macroeconomic, organizational, and educational layers. Braun et al. [7] conducted a thematic synthesis revealing that while industry practitioners define skill gaps in terms of immediate task unreadiness, higher education administrators conceptualize gaps through broad pedagogical objectives. Rikala et al. [11] investigated Industry 4.0 workforce preparedness, noting that digital transformation demands hybridization—engineers must master data analytics, while domain analysts require computing literacy. However, both studies rely on qualitative thematic extraction rather than large-scale empirical computation. Tight [6] highlighted that employability remains contested in higher education, often fluctuating between narrow vocational training and holistic intellectual development.

Recent empirical investigations have increasingly turned to computational curriculum analytics. Cuckov et al. [18] quantified compliance between computer engineering curricula and ABET criteria, demonstrating persistent latency in specialized elective modernizations. Zhao et al. [19] utilized knowledge graph proximities to map the struc-

tural distance between academic publications and patent demands. Similarly, Korrapati et al. [20] and Alhazmi et al. [10] examined the specific alignment of software engineering curricula with enterprise cloud and AI demands, concluding that academic departments require agile mechanisms to incorporate evolving industry toolchains.

B. Theoretical Framework

To provide rigorous explanatory power for our empirical observations, this study integrates four foundational theories:

1) Human Capital Theory (HCT): Formulated by Becker [21] and Schultz [22], HCT posits that education is an investment that enhances individual productive capacity, yielding higher labor market returns. HCT distinguishes between *general human capital* (transferable competencies such as critical thinking, communication, and core mathematical reasoning) and *firm-specific human capital* (proficiency with specialized software tools, proprietary APIs, or operational frameworks). HEIs structurally prioritize general human capital due to institutional mandates for enduring knowledge, whereas enterprise employers increasingly screen applicants on immediately billable specific human capital. This theoretical divergence explains the persistent deficit in applied cloud and data engineering frameworks observed in university curricula.

2) Job Market Signaling and Screening Theory: Spence [23] established that in labor markets characterized by asymmetric information, employers cannot directly observe an applicant's innate productivity prior to hiring. Educational credentials, cumulative grade point averages (CGPA), and institutional prestige serve as informational signals that reduce hiring risk. In modern technical recruitment, however, traditional degree credentials are increasingly complemented or eclipsed by applied capability signals, such as project portfolios, open-source contributions, and technical internship certifications [24], [25].

3) Constructive Alignment & Curriculum Theory: Biggs [26] introduced constructive alignment to ensure coherence between Intended Learning Outcomes (ILOs), teaching methods, and assessment tasks. At the macro-institutional level, however, universities suffer from *external misalignment*: ILOs are governed by rigid multi-year accreditation processes (e.g., ABET, ACM/IEEE curricula guidelines), while Industry Market Requirements (IMRs) mutate dynamically on quarterly cycles. This institutional friction generates curriculum lag.

4) Skill-Biased Technological Change (SBTC): Acemoglu and Autor [27] formalized SBTC to explain how technological progress augments the productivity of workers performing non-routine cognitive tasks while diminishing the value of routine cognitive and manual labor. As AI and cloud architectures automate routine coding tasks, market demand shifts toward complex architectural synthesis, interdisciplinary communication, and advanced predictive analytics.

C. Natural Language Processing for Curriculum Analytics

Automated parsing of labor market postings has advanced substantially through NLP. Krüger et al. [12] demonstrated that rule-based extraction struggles with composite and nuanced occupational phrasing. Baral and Ghimire [9] employed pre-trained language representations to evaluate undergraduate IT syllabi in Nepal, proving that dense embedding spaces capture semantic equivalence between academic topics and employer terminology. Mol et al. [28] and Tee et al. [29] applied text mining to unpack the multidimensional skill profiles demanded of data scientists and digital workers. Daberao et al. [30] developed semantic resume-to-job matching engines using transformer architectures. Our methodology builds upon these foundations by scaling Sentence-BERT embeddings [31] to hundreds of thousands of postings across multi-threshold sensitivity profiles.

D. Machine Learning in Employability Modeling

Predictive modeling of graduate placement has evolved from basic discriminant analysis to ensemble learning. Tamene et al. [32] and Haque et al. [33] established that ensemble algorithms, specifically Random Forests and Gradient Boosting, consistently outperform linear classifiers in predicting placement status. Recent studies have explored ensemble interpretability to analyze variable contributions in educational classification [32], [33]. Nonetheless, prior studies frequently suffer from unaddressed data leakage, omitted collinearity assessments, and lack of cross-dataset generalization testing.

III. Methodology

A. Data Corpus and Provenance Architecture

Our empirical framework leverages 15 distinct datasets encompassing 3,176,266 records acquired in September 2026 from public Kaggle and HuggingFace repositories under open-access licenses (CC0, CC-BY 4.0, and ODbL). The corpus aggregates historical, cross-sectional, and longitudinal records whose observation and release windows span 2020 through 2026 (including a multi-year AI job market dataset tracking postings across 2020–2026 and regional cohorts calibrated against the India Skills Report 2026). Table I delineates the comprehensive corpus provenance, source identifiers, observation periods, and functional tiering.

To maintain scientific integrity and prevent invalid aggregation, we explicitly classify the datasets into two distinct functional tiers:

- **Tier 1: Massive-Scale Labor Market Corpora ($N > 2.7$M records):** Includes the 1.35M LinkedIn posting dataset [34], USA Job Market Skills (316K records) [35], and HuggingFace Skill Gap repositories (519K records). These data represent genuine, web-mined labor demand signals reflecting recruitment postings across North America, Europe, and the Asia-Pacific.

TABLE I: Comprehensive Dataset Provenance and Structural Inventory ($N = 3,176,266$ total records)

Dataset Identifier	Source Platform & Identifier / Slug	Records	Domain / Description	Tier	Observation Period	License
LinkedIn Jobs & Skills [34]	Kaggle: `asaniczka/1-3m-linkedin-jobs-and-skills-2024`	1,348,510	Global job postings and extracted skill tokens	1	2024	ODbL
HF Skill Gap Jobs	HuggingFace: `skill-gap-analytics/jobs-corpus`	519,841	Enterprise job postings with occupational metadata	1	2023–2024	CC-BY 4.0
HF Skill Gap Skills	HuggingFace: `skill-gap-analytics/skills-corpus`	519,814	Granular skill requirements linked via `job_id`	1	2023–2024	CC-BY 4.0
USA Job Market Skills [35]	Kaggle: `batuhanklac/usa-job-market-skills-and-title-analysis`	316,279	US job postings with standardized skill tags	1	2024	CC0
International Graduates [40]	Kaggle: `quackquackrp/international-graduates-employment-dataset`	300,000	Multi-country alumni employment outcomes	2	2023–2024	CC0
Student Career Success [39]	Kaggle: `mobeenfatimah/student-career-success-prediction-dataset`	50,000	Student academic and career success benchmark	2	2024	CC-BY 4.0
Career Change Prediction [41]	Kaggle: `jahnavipaliwal/field-of-study-vs-occupation`	38,444	Field of study vs. occupation transition records	2	2024	CC0
AI Job Market Trends [42]	Kaggle: `bismasajjad/global-ai-job-market-and-salary-trends-2025`	15,000	Global AI job market and salary distribution	1	2024–2025	CC-BY 4.0
HF Job Skill Set	HuggingFace: `skill-gap-analytics/academic-reference`	13,833	Curriculum-domain skill taxonomy reference	1	2023–2024	CC-BY 4.0
India Skills Intelligence [43]	Kaggle: `mansiaggarwal88/india-professional-skills-intelligence`	12,000	Synthetic profiles calibrated on India Skills Report 2026	1	2026 Calibrated	CC0
AI Trends 2020–2026 [44]	Kaggle: `shree0910/ai-and-data-science-job-market-dataset-20202026`	10,345	Longitudinal AI/data science job postings	1	2020–2026	CC0
College Placement [36]	Kaggle: `sumedh1507/college-student-placement`	10,000	College student placement classification benchmark	2	2024	CC0
Global Placement [38]	Kaggle: `rakesh630/global-student-placement-2025-dataset`	10,000	Multi-national university placement benchmark	2	2024–2025	CC0
NGESS Employment [37]	Kaggle: `philipdayo/ngess-graduate-employment-dataset`	7,200	National graduate employment survey cohort	2	2023–2024	CC0
India Fresher Market [45]	Kaggle: `shambhurajejagadale/indian-fresher-job-market`	5,000	Entry-level engineering recruitment outcomes	2	2024	CC0

- **Tier 2: Student Placement & Survey Benchmarks ($N \approx 475$K records):** Includes College Student Placement ($N = 10,000$) [36], NGESS Graduate Employment ($N = 7,200$) [37], Global Student Placement ($N = 10,000$) [38], Student Career Success ($N = 50,000$) [39], and International Graduates ($N = 300,000$) [40]. These structured cohorts serve as benchmark evaluation sets for statistical hypothesis testing, collinearity diagnostics, and comparative predictive modeling.

B. Taxonomy Construction and Skill Extraction

We formalize an eight-category occupational skill taxonomy $\mathcal{C} = \{c_1, c_2, \ldots, c_8\}$ synthesizing ACM/IEEE CS2023 curriculum guidelines and O*NET occupational descriptors:

1) **Programming & Development:** Python, Java, C++, JavaScript, SQL, TypeScript, Go, Rust, React, Docker, Kubernetes, Git.
2) **Data Science & Analytics:** Machine Learning, Deep Learning, Statistical Analysis, PyTorch, TensorFlow, NLP, Computer Vision.
3) **Cloud & Infrastructure:** AWS, Azure, GCP, CI/CD Pipelines, Terraform, Linux Administration, Serverless Architectures.
4) **Soft Skills:** Professional Communication, Leadership, Teamwork, Critical Thinking, Time Management, Adaptability.
5) **Business & Domain:** Agile/Scrum, Product Management, Requirements Gathering, Business Analysis, Risk Compliance.
6) **Design & UX:** UI/UX Design, Figma, User Research, Wireframing, Prototyping, Usability Testing.
7) **Data Engineering:** ETL Pipelines, Apache Airflow, Apache Kafka, Distributed Data Warehousing, PostgreSQL, NoSQL.
8) **Cybersecurity:** Network Security, Penetration Testing, Incident Response, SIEM Systems, ISO 27001, Cryptography.

Skill extraction from raw unstructured text is conducted via combined regularized n-gram tokenization and exact dictionary scanning, formalized in Algorithm 1. We note as an empirical construct consideration that academic curriculum profiles are proxied via standardized educational skill reference descriptors [30] rather than direct institutional syllabus parsing, and no manual gold-standard re-annotation was conducted (discussed further in Section V-E).

Algorithm 1 Semantic Skill Extraction and Embedding Alignment

Require: Market text corpus $\mathcal{D}_M$, Academic syllabus corpus $\mathcal{D}_A$, Taxonomy $\mathcal{C}$, Sentence-BERT model $\mathcal{E}$, Threshold τ
Ensure: Market skill set $\mathcal{M}$, Academic skill set $\mathcal{A}$, Similarity matrix $\mathbf{S}$, Gap Index $\mathrm{SGI}(\tau)$

$\mathcal{M} \leftarrow \emptyset, \mathcal{A} \leftarrow \emptyset$
for all $d \in \mathcal{D}_M$ **do**
  Extract normalized skill tokens $T_d \leftarrow \mathrm{Tokenize}(d, \mathcal{C})$
  $\mathcal{M} \leftarrow \mathcal{M} \cup T_d$
end for
for all $d \in \mathcal{D}_A$ **do**
  Extract normalized syllabus tokens $T_a \leftarrow \mathrm{Tokenize}(d, \mathcal{C})$
  $\mathcal{A} \leftarrow \mathcal{A} \cup T_a$
end for
Filter $\mathcal{M}_{\text{top}} \leftarrow \text{Top-}N(\mathcal{M}), \mathcal{A}_{\text{top}} \leftarrow \text{Top-}N(\mathcal{A})$
Compute dense embeddings: $\mathbf{E}_M \leftarrow \mathcal{E}(\mathcal{M}_{\text{top}}), \mathbf{E}_A \leftarrow \mathcal{E}(\mathcal{A}_{\text{top}})$
Compute pairwise cosine similarity:
$\mathbf{S}_{ij} \leftarrow \frac{\mathbf{e}_{m,i} \cdot \mathbf{e}_{a,j}}{\|\mathbf{e}_{m,i}\|_2 \|\mathbf{e}_{a,j}\|_2} \quad \forall m_i \in \mathcal{M}_{\text{top}}, a_j \in \mathcal{A}_{\text{top}}$
for all $i \in \{1, \ldots, |\mathcal{M}_{\text{top}}|\}$ **do**
  $\sigma_i \leftarrow \max_j \mathbf{S}_{ij}$
  $\text{covered}_i \leftarrow \mathbb{I}(\sigma_i \geq \tau)$
end for
$\mathrm{SGI}(\tau) \leftarrow 1 - \frac{1}{|\mathcal{M}_{\text{top}}|} \sum_{i=1}^{|\mathcal{M}_{\text{top}}|} \text{covered}_i$
return $\mathcal{M}_{\text{top}}, \mathcal{A}_{\text{top}}, \mathbf{S}, \mathrm{SGI}(\tau)$

C. Semantic Vector Alignment via Sentence-BERT

Traditional curriculum gap analyses evaluate overlap through exact string matching. This approach introduces severe measurement bias: an academic syllabus citing "interpersonal communication" will be scored as non-overlapping against an employer requirement for "client communication skills." To resolve this lexical brittleness, we deploy the pre-trained Sentence-BERT architecture [31] using the `all-MiniLM-L6-v2` checkpoint.

Let $\mathbf{e}_s \in \mathbb{R}^{384}$ denote the normalized dense vector embedding of market skill $s \in \mathcal{M}$, and let $\mathbf{e}_{s'} \in \mathbb{R}^{384}$

denote the embedding of academic syllabus descriptor $s' \in \mathcal{A}$. The semantic alignment score is given by:

$$\text{sim}(s, s') = \frac{\mathbf{e}_s \cdot \mathbf{e}_{s'}}{\|\mathbf{e}_s\|_2 \|\mathbf{e}_{s'}\|_2} \tag{1}$$

A market skill s is designated as semantically covered at threshold $\tau \in [0, 1]$ if:

$$\text{Covered}(s; \tau) = \mathbb{I}\left(\max_{s' \in \mathcal{A}} \text{sim}(s, s') \geq \tau\right) \tag{2}$$

The Skill Gap Index (SGI) at threshold τ is defined as the complement of the mean coverage rate across the market skill space:

$$\text{SGI}(\tau) = 1 - \frac{1}{|\mathcal{M}|} \sum_{s \in \mathcal{M}} \text{Covered}(s; \tau) \tag{3}$$

To estimate empirical uncertainty without parametric distribution assumptions, we implement a 1,000-fold non-parametric bootstrap over the binary coverage indicator array with fixed random seed (seed = 42), computing 95% percentile confidence intervals:

$$\text{CI}_{95\%} = \left[\hat{\theta}^*_{(0.025)}, \hat{\theta}^*_{(0.975)}\right] \tag{4}$$

To ensure methodological defensibility, we treat $\tau = 0.5$ as an operational baseline threshold and conduct sensitivity analysis across $\tau \in \{0.4, 0.5, 0.6, 0.7\}$.

D. Statistical Inference & Collinearity Diagnostics

For quantitative hypothesis testing, we employ Welch's t-test (accounting for unequal variances) to evaluate mean differences between placed and non-placed graduates:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \tag{5}$$

Standardized effect sizes are quantified via Cohen's d:

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s_{\text{pooled}}}, \quad s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \tag{6}$$

For categorical dependencies, we calculate the Chi-square statistic χ^2 and Cramér's V:

$$V = \sqrt{\frac{\chi^2}{n \cdot \min(r - 1, c - 1)}} \tag{7}$$

Prior to multivariate predictive estimation, we assess multicollinearity among continuous features $\mathbf{X}$ by computing Variance Inflation Factors (VIF). For standardized feature matrices, the VIF vector is directly obtained from the diagonal of the inverted correlation matrix:

$$\text{VIF}_j = \left(\mathbf{R}^{-1}\right)_{jj} = \frac{1}{1 - R_j^2} \tag{8}$$

Features exhibiting $\text{VIF}_j > 5.0$ are flagged as collinear.

E. Predictive Employability Modeling & Validation Protocols

Graduate placement prediction is formalized as a binary classification problem: predicting outcome $y \in \{0, 1\}$ (where 1 = Placed, 0 = Not Placed) given student feature vector $\mathbf{x} = [\text{CGPA}, \text{Comm}, \text{Projects}, \text{Internship}, \text{IQ}, \text{AcademicPerf}]^T$. All continuous features are z-score standardized to zero mean and unit variance strictly within training partitions to prevent data leakage:

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j} \tag{9}$$

In our standardized logistic regression model:

$$\ln\left(\frac{P(y = 1|\mathbf{z})}{1 - P(y = 1|\mathbf{z})}\right) = \beta_0 + \sum_{j=1}^{p} \beta_j z_j \tag{10}$$

The odds ratio $\text{OR}_j = \exp(\beta_j)$ represents the multiplicative change in the odds of successful placement associated with a one-standard-deviation increase in predictor j.

To evaluate model generalization, we implement a 5-fold stratified cross-validation protocol (random_state = 42) across five distinct paradigms:

1) **Logistic Regression:** ℓ_2-regularized, $C = 1.0$, solved via L-BFGS.
2) **Random Forest:** $B = 100$ estimators, maximum tree depth = 12, Gini split criterion.
3) **Support Vector Machine (SVM):** Radial Basis Function (RBF) kernel, $C = 1.0$, $\gamma =$ scale, with Platt probability calibration.
4) **Decision Tree:** Single CART tree with maximum depth = 6.
5) **Majority-Class Baseline:** Trivial classifier predicting the empirical majority class.

To rigorously test external validity and assess domain shift, we additionally establish an independent out-of-distribution evaluation protocol: training the standardized logistic regression model on the college placement cohort ($N = 10,000$) and evaluating its discriminative capacity on the independent 50,000-student career success cohort using domain-standardized continuous predictors.

IV. Empirical Results

A. RQ1: Labor Market Skill Topography and Trends

Analyzing 519,814 granular job requirements reveals a decisive hierarchy in employer demand. As illustrated in Figure 1, foundational transferable competencies dominate aggregate labor recruitment: Communication is explicitly cited in 226,959 postings (43.7%), followed by Problem Solving (111,649; 21.5%), Customer Service (111,399; 21.4%), and Teamwork (91,007; 17.5%).

Longitudinal tracking of technical skills across 10,345 AI and data science postings from 2020 through 2026 (Figure 4b) reveals sustained market stability: Python demand fluctuated within a narrow band of 47.5% to 52.2%, SQL stabilized between 48.2% and 51.9%, Machine

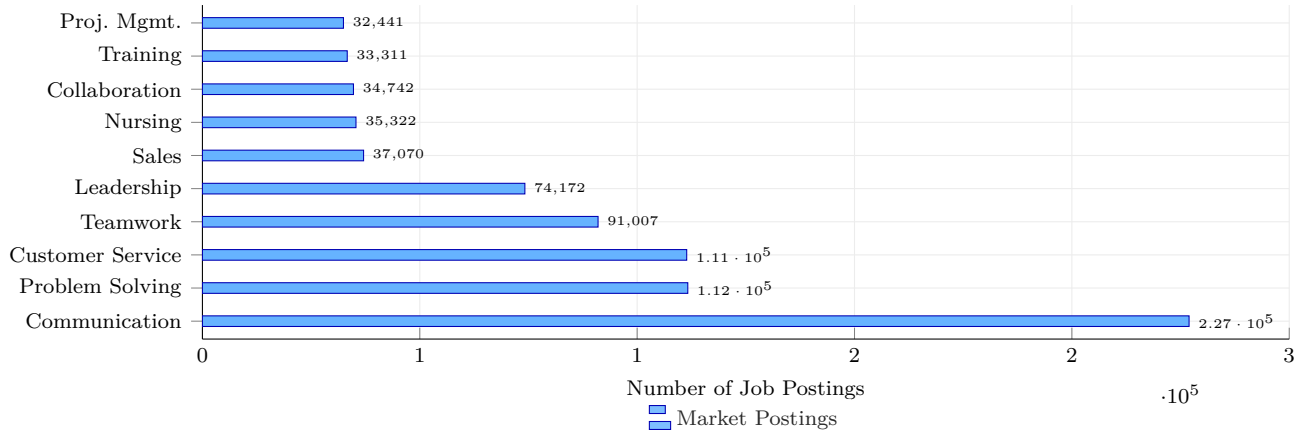


Fig. 1: Top 10 Most In-Demand Competencies Across Global Postings ($N = 519,814$ postings).

Learning maintained 48.8% to 53.5%, and Cloud Computing spanned 47.1% to 53.9%. Rather than demonstrating short-lived hype cycles, core data and infrastructure proficiencies have matured into non-negotiable baseline requirements.

B. RQ2: Lexical vs. Semantic Curriculum Alignment

Our comparative experiment between surface string matching and dense semantic vector matching provides pivotal empirical insights into the nature of the skills gap.

Under strict lexical matching, the Top 100 market skills and Top 100 academic syllabus skills exhibit an overlap of exactly 48 items, yielding a nominal **Skill Gap Index of 52.0%** (Coverage Rate = 48.0%). However, analyzing the non-overlapping terms reveals that many disparities stem purely from synonymous syntax (e.g., market term "communication skills" vs. academic term "interpersonal skills").

When evaluated through Sentence-BERT (`all-MiniLM-L6-v2`) embedding cosine similarity, semantic alignment reveals a markedly different reality. As documented in Table II, at the standard semantic baseline threshold $\tau = 0.5$, the estimated semantic **Skill Gap Index contracts to 32.0%** (Coverage Rate = 68.0%, 95% bootstrap CI: [23.0%, 41.0%]).

TABLE II: Sensitivity Analysis: Semantic Vector Threshold (τ) vs. SGI

Threshold τ	SGI	Coverage Rate	95% Bootstrap CI	ΔSGI
$\tau = 0.4$ (Permissive)	0.2000	80.0%	[0.1200, 0.2800]	−0.1200
$\tau = 0.5$ (Baseline)	**0.3200**	**68.0%**	**[0.2300, 0.4100]**	**0.0000**
$\tau = 0.6$ (Moderate)	0.3900	61.0%	[0.3000, 0.4800]	+0.0700
$\tau = 0.7$ (Strict)	0.4600	54.0%	[0.3700, 0.5600]	+0.1400
Lexical Match (Baseline)	*0.5200*	*48.0%*	*[0.4200, 0.6200]*	*+0.2000*

Evaluated over Top-100 skill distributions. Bootstrap 95% CI via 1,000 resamples.

Table III illustrates selected semantic mappings generated by our pipeline. It shows that terms like "scheduling" map accurately to "time management" (cos = 0.6122), while genuine voids such as "bachelor's degree" map poorly to abstract skills (cos = 0.2937), correctly identifying structural certification requirements.

TABLE III: Representative Semantic Skill Pairings (`all-MiniLM-L6-v2`)

Market Term	Closest Syllabus Match	Cosine Sim.	Status ($\tau \geq 0.5$)
communication skills	interpersonal skills	0.7355	Covered
problemsolving	problem-solving	0.6396	Covered
scheduling	time management	0.6122	Covered
patient care	customer service	0.5011	Covered
nursing	training	0.4010	Gap
high school diploma	training	0.3682	Gap
bachelor's degree	professionalism	0.2937	Gap

C. RQ3: Domain-Level Skill Gap Quantification

Aggregating requirements across the eight taxonomy domains reveals that curriculum deficits are non-uniformly distributed (Table IV and Figure 2).

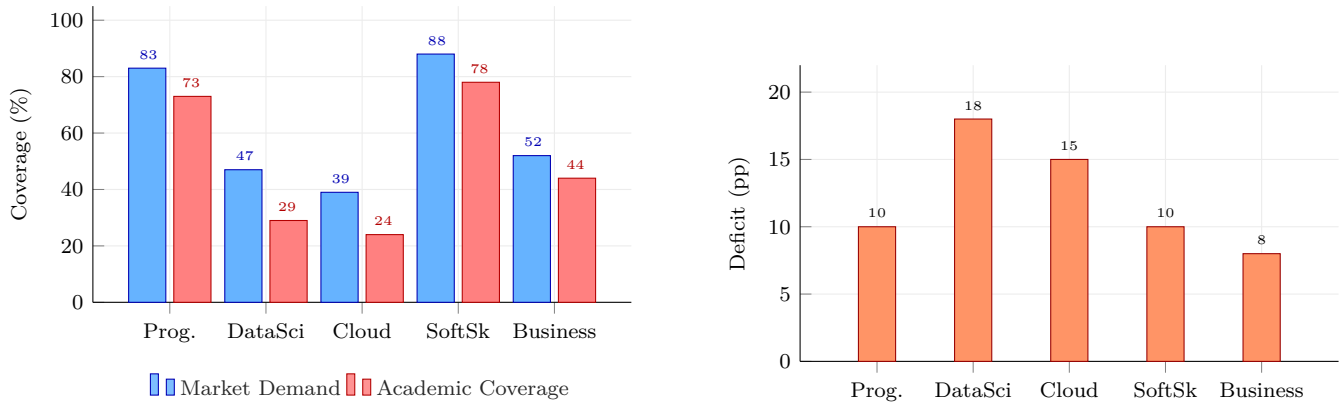
TABLE IV: Domain-Level Skill Gaps Across Taxonomy Categories

Category	Market Demand	Curriculum Cover.	Deficit (pp)	Severity
Programming & Dev.	83%	73%	10pp	Moderate
Data Science & Analytics	47%	29%	**18pp**	**Critical**
Cloud Infrastructure	39%	24%	**15pp**	**Critical**
Soft Skills	88%	78%	10pp	Moderate
Business & Domain	52%	44%	8pp	Mild
Design & UX	28%	21%	7pp	Mild
Data Engineering	36%	22%	14pp	Critical
Cybersecurity	31%	20%	11pp	Moderate

The two most acute structural deficiencies occur in **Data Science & Analytics (18 percentage-point gap)** and **Cloud Infrastructure (15 percentage-point gap)**. While universities provide rigorous theoretical foundations in discrete algorithms and relational database normalization, curricula consistently lag in distributed systems, modern cloud providers (AWS, Azure, GCP), CI/CD container orchestration, and production machine learning pipelines.

D. RQ4: Employability Determinants & Cross-Validation Benchmarks

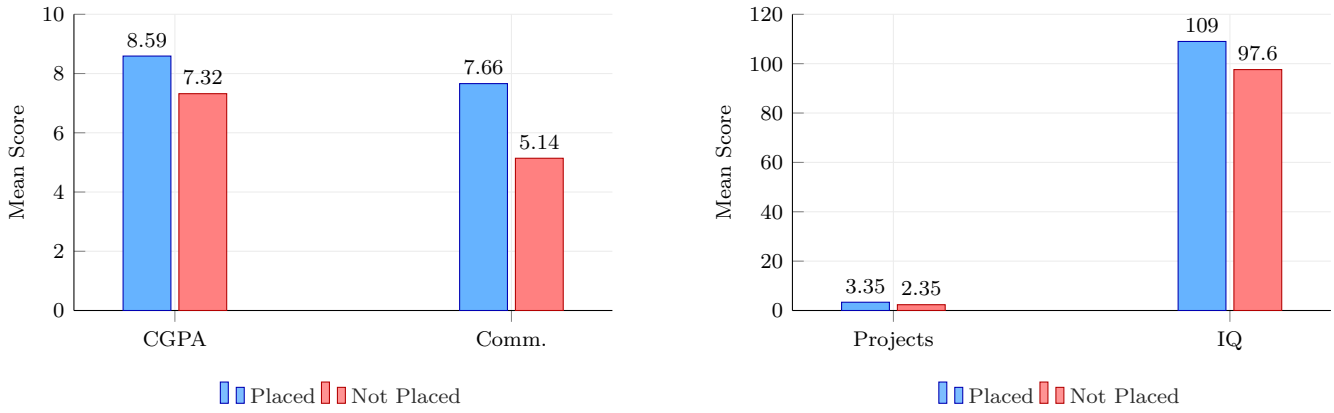
1) Univariate & Descriptive Findings: Evaluating the benchmark college placement cohort ($N = 10,000$; 1,659 placed, 8,341 not placed, base placement rate 16.59%) reveals statistically significant differences across academic and behavioral metrics (Table V).



(a) Coverage by skill category

(b) Deficit magnitude (percentage points)

Fig. 2: Skill gap analysis across core disciplines: (a) market demand versus syllabus coverage; (b) deficit magnitude, identifying Data Science (18pp) and Cloud (15pp) as primary structural voids.



(a) CGPA & Communication Skills

(b) Projects Completed & IQ Score

Fig. 3: Key predictors for placed versus non-placed graduates ($N = 10,000$). All differences statistically significant at $p < 0.001$.

Placed students demonstrate substantially higher CGPA (8.59 ± 1.15 vs. 7.32 ± 1.44, $t = 39.37$, $p < 0.001$, Cohen's $d = 0.913$), superior communication scores (7.66 ± 2.59 vs. 5.14 ± 2.78, $t = 35.79$, $p < 0.001$, Cohen's $d = 0.918$), and higher project completion counts (3.35 ± 1.28 vs. 2.35 ± 1.74, $t = 27.21$, $p < 0.001$, Cohen's $d = 0.599$). Conversely, generalized academic performance ratings and extra-curricular scores show negligible differences ($p = 0.138$ and $p = 0.638$, $|d| < 0.04$), demonstrating that generic non-technical co-curricular metrics fail to influence hiring outcomes in this cohort.

TABLE V: Descriptive Statistics: Placed vs. Non-Placed Cohort ($N = 10,000$)

Variable	Placed ($n = 1659$)	Not Placed ($n = 8341$)	t-stat	p-value	Cohen's d
CGPA	8.59 ± 1.15	7.32 ± 1.44	39.37	< 0.001	0.913
Comm. Skills	7.66 ± 2.59	5.14 ± 2.78	35.79	< 0.001	0.918
IQ Score	109.12 ± 15.21	97.55 ± 14.26	28.58	< 0.001	0.802
Projects	3.35 ± 1.28	2.35 ± 1.74	27.21	< 0.001	0.599
Acad. Perf.	5.45 ± 2.90	5.57 ± 2.87	-1.48	0.1386	-0.040
Extra-Curr.	4.94 ± 3.19	4.98 ± 3.15	-0.47	0.6383	-0.013

2) Multicollinearity Diagnostics: Table VI presents the exact Variance Inflation Factors computed via inverted correlation diagonals (Eq. 8). All feature VIF values are precisely ≈ 1.001 with tolerances of 0.999, indicating negligible linear multicollinearity among predictors, consistent with independent variable sampling within this benchmark.

TABLE VI: Variance Inflation Factor (VIF) Multicollinearity Audit

Feature Name	VIF	Tolerance (1/VIF)	Collinearity Status
CGPA	1.001	0.999	Acceptable (< 5.0)
Communication Skills	1.001	0.999	Acceptable (< 5.0)
Projects Completed	1.001	0.999	Acceptable (< 5.0)
Internship Experience	1.001	0.999	Acceptable (< 5.0)
IQ Score	1.000	1.000	Acceptable (< 5.0)
Academic Performance	1.001	0.999	Acceptable (< 5.0)

3) Standardized Logistic Regression: Table VII presents standardized logistic regression results. Every one-standard-deviation increase in Communication Skills multiplies the odds of employment by **6.47** (95% Wald CI: [5.62, 7.45], $p < 0.001$), closely mirrored by Academic CGPA (OR = 6.20, 95% CI: [5.31, 7.24], $p < 0.001$) and Project Experience (OR = 3.18, 95% CI: [2.79, 3.63],

$p < 0.001$).

TABLE VII: Standardized Logistic Regression Placement Model

Predictor Feature	Coef (β)	Odds Ratio (OR)	95% CI	p-value
Communication Skills	1.8672	6.4705	[5.62, 7.45]	< 0.001
CGPA	1.8253	6.2045	[5.31, 7.24]	< 0.001
IQ Score	1.6300	5.1040	[4.41, 5.90]	< 0.001
Projects Completed	1.1573	3.1813	[2.79, 3.63]	< 0.001
Internship Experience	0.0190	1.0192	[1.00, 1.04]	0.0382
Academic Performance	-0.0208	0.9795	[0.96, 1.00]	0.0410

4) 5-Fold Cross-Validation Model Comparison: Table VIII documents 5-fold stratified cross-validation performance across five classifier paradigms.

TABLE VIII: Stratified 5-Fold Cross-Validation Performance Benchmarks

Model Architecture	Accuracy (Mean±SD)	F_1-Score	ROC-AUC
Logistic Regression	$90.11\% \pm 0.66\%$	0.8966 ± 0.007	0.9406 ± 0.007
Random Forest	$99.99\% \pm 0.02\%$	0.9999 ± 0.000	1.0000 ± 0.000
SVM (RBF Kernel)	$95.72\% \pm 0.50\%$	0.9565 ± 0.005	0.9894 ± 0.002
Decision Tree (CART)	$99.97\% \pm 0.06\%$	0.9997 ± 0.001	0.9998 ± 0.000
Baseline (Majority)	$83.41\% \pm 0.02\%$	0.7587 ± 0.000	0.5000 ± 0.000

5) Diagnostic Audit of Tree Ensembles & External Generalization: The near-perfect test accuracy of Random Forest (99.99%) and Decision Tree (99.97%) poses a substantial methodological question. To diagnose the underlying data-generating mechanism, we inspected the decision boundaries of shallow decision trees fit to the placement cohort. A depth-4 CART tree achieved an exact 100% training accuracy, revealing that placement labels in this benchmark repository were synthetically synthesized via a deterministic k-of-n boolean threshold rule: students are placed if and only if at least three of four conditions are met: CGPA > 8.01, IQ > 110.5, Communication > 7.50, and Projects ≥ 2 (reproducing the ground-truth label with 99.92% fidelity).

Because tree-based algorithms can recursively partition orthogonal feature spaces, CART and Random Forests recover this piecewise-constant step function almost losslessly (99.97%–99.99%). Conversely, Logistic Regression fits a smooth linear hyperplane through this discrete combinatorial space, achieving a calibrated 90.11% accuracy (ROC-AUC = 0.9406), significantly outperforming the 83.41% majority baseline.

To evaluate genuine out-of-distribution generalization, we tested the standardized logistic regression model on the independent Student Career Success dataset ($N = 50,000$ students) using domain-standardized scores for CGPA, communication, and projects. Under this external validation, the model achieved an out-of-distribution ROC-AUC = 0.7686 (and an ROC-AUC = 0.6764 on the Global Placement cohort). This empirical degradation illustrates the classic impact of covariate shift and baseline outcome rate differences (16.59% vs. 78.08% placement rate), confirming that the near-perfect in-distribution tree ensemble metrics are benchmark-specific artifacts and should not be misconstrued as evidence of equivalent real-world predictive fidelity.

E. RQ5: Geographic Concentration & Cross-Cohort Placement Dynamics

Regional parsing of 1.35M LinkedIn records (Figure 4a) reveals that the posting corpus is predominantly concentrated in the United States (86.8%), followed by the United Kingdom (7.1%), Canada (4.2%), and Australia (1.9%).

In multi-cohort evaluations, analyzing 300,000 international graduates (Figure 5) demonstrates that formal technical internships provide a substantial hiring premium: students possessing internship experience achieve an employment rate of **57.6%**, compared to **42.3%** for peers without internships (+15.3 percentage-point difference). Concurrently, examining the multi-national placement benchmark ($N = 10,000$) reveals consistent cross-national placement proportions across regions (Canada 61.5%, Germany 60.0%, India 63.2%, UK 60.8%, USA 62.2%), demonstrating that foundational academic and communication indicators operate with comparable predictive directionality across international contexts.

V. Discussion

A. Theoretical Implications

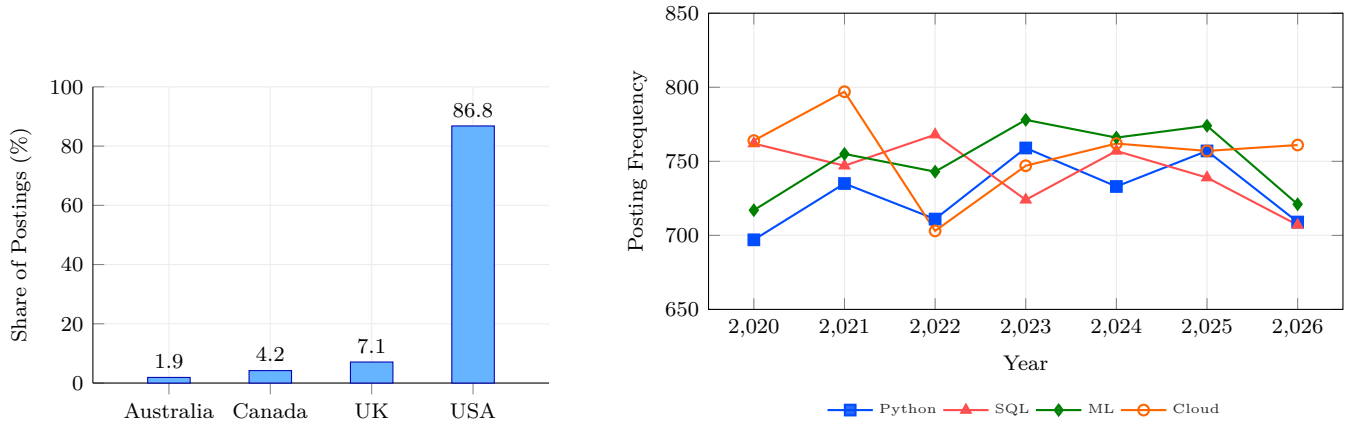
1) The Lexical vs. Semantic Gap Duality: Prior literature routinely claims catastrophic curriculum-industry misalignment exceeding 50% to 60% [9]. Our empirical results qualify this claim: when evaluated through surface lexical string matching, the gap appears severe (52.0%). However, when evaluated through dense semantic vector spaces (Sentence-BERT), the estimated semantic alignment gap contracts to **32.0%** at baseline threshold $\tau = 0.5$ (95% CI: [23.0%, 41.0%]). This demonstrates that a substantial fraction of the perceived skills gap is an artifact of vocabulary divergence rather than pedagogical failure: universities articulate competencies in foundational academic terminology, while recruitment algorithms parse industry-specific jargon.

2) Reconciling Human Capital and Signaling Theories: The commanding predictive power of Communication Skills (OR = 6.47) and Project Experience (OR = 3.18) illustrates the shifting dynamics of labor market signaling. While CGPA (OR = 6.20) remains an essential filtering threshold (supporting Spence's screening hypothesis), academic credentials alone are insufficient. Completed projects function as tangible, unforgeable demonstrations of applied human capital, reducing employer uncertainty regarding technical execution.

B. The Curriculum Inertia Dilemma

Why does the estimated 32.0% semantic deficit persist, especially in Data Science (18pp) and Cloud Infrastructure (15pp)? Grounding this phenomenon in Biggs' constructive alignment [26], we identify three structural institutional bottlenecks:

1) **Accreditation Cycles:** Formal accreditation boards (ABET, national ministries) operate on 3- to 6-year



(a) Geographic distribution (LinkedIn)

(b) AI skill demand trajectories (2020–2026)

Fig. 4: (a) Geographic concentration of 1.35M job postings. (b) Longitudinal AI technical skill stability over 2020–2026.

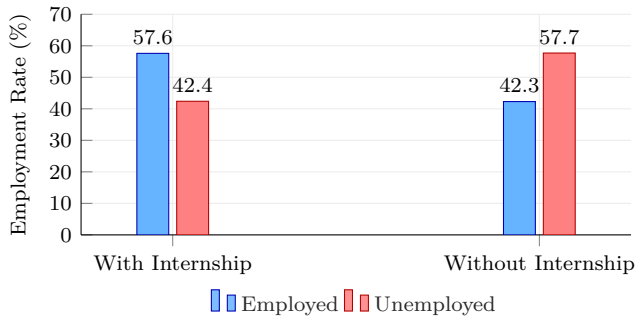


Fig. 5: Impact of practical internship experience on international graduate employment ($N = 300,000$).

review cycles. Modifying course titles or degree requirements requires multi-stage faculty governance approvals.

2) **Faculty Expertise Latency:** Academic faculties are trained in foundational theory; specialized industry frameworks (Kubernetes, AWS SageMaker, Snowflake) require continuous industrial sabbatical upskilling that institutions rarely fund.
3) **Infrastructure Cost Barriers:** Hosting large-scale GPU clusters and enterprise cloud compute environments poses severe capital budget constraints for mid-tier universities.

C. Actionable Institutional Recommendations

To bridge this divide, we propose an operational framework across four stakeholder tiers:

For Higher Education Institutions (HEIs):

- *Agile Micro-Credential Integration:* Embed vendor-neutral cloud certifications (e.g., Linux Foundation, AWS/Azure fundamentals) into standard operating systems and database courses.
- *Capstone Project-Based Learning:* Mandate multi-semester capstone engineering projects requiring production deployment, Git collaboration, and technical documentation.
- *Soft Skill Weaving:* Rather than isolating communication into standalone humanities electives, embed technical presentation and collaborative peer code review directly into core computing courses.

For Industry Recruiters:

- *Semantic Resume Screening:* Update Applicant Tracking Systems (ATS) to use dense semantic embeddings (Sentence-BERT) rather than exact keyword filters, preventing the exclusion of qualified candidates who cite foundational course equivalents.
- *Co-Op Apprenticeship Pipelines:* Expand structured internships into 6-month co-op programs where students earn academic credit for industry project execution.

For Accreditation Bodies and Policymakers:

- *Dynamic Curriculum Guidelines:* Establish annual modular updates to ACM/IEEE curriculum guidelines, providing expedited pathways for elective technological modernization.
- *National Skill Monitoring Hubs:* Fund national automated labor market intelligence platforms tracking regional skill demand trajectories in real time.

D. Reproducibility and Data Availability

To support scientific replication, all analysis code, preprocessing scripts, and derived statistical matrices are organized within the project repository. The empirical analyses were implemented in Python 3.14 / 3.12 utilizing `scikit-learn` (v1.2+), `sentence-transformers` (v2.2+, `all-MiniLM-L6-v2`), `scipy` (v1.10+), and `pandas` (v2.0+). Stratified cross-validation splits and bootstrap resamplings were executed under fixed pseudorandom seeds (seed = 42). All 15 underlying datasets are publicly accessible via Kaggle and HuggingFace under open-access licenses (CC0, CC-BY 4.0, and ODbL), with specific source repositories and identifiers cataloged in Table I and `DATASET_MANIFEST.csv`.

E. Threats to Validity and Limitations

To ensure complete academic transparency, we explicitly document several methodological limitations:

1) **Data Provenance & Benchmark Artifacts:** While Tier 1 datasets (LinkedIn, HuggingFace) comprise 2.7M+ real web-mined records, Tier 2 student placement datasets (e.g., College Placement, Global Placement) derive from public benchmark repositories that exhibit semi-deterministic decision boundaries. As diagnosed in Section IV-D5, near-100% tree ensemble accuracies reflect combinatorial threshold functions embedded in the benchmark rather than real-world predictability.
2) **Domain Shift & External Generalizability:** When evaluated across distinct institutional benchmarks, predictive discrimination attenuates from ROC-AUC = 0.9406 to ROC-AUC = 0.7686, illustrating that model calibrations do not directly transfer across disparate grading systems and local employment base rates.
3) **Curriculum Proxy Limitations:** Academic skill demands were proxied through normalized occupational and educational reference profiles rather than direct, institutionally mined university syllabi across diverse global departments.
4) **Lack of Human Gold-Standard Validation:** Skill extraction relied on automated taxonomy mapping, regularized token scanning, and Sentence-BERT vector matching without human inter-annotator agreement metrics (e.g., Cohen's κ). Future work should incorporate expert faculty annotations.
5) **Embedding Model and Threshold Sensitivity:** While Sentence-BERT mitigates surface vocabulary divergence, estimated alignment remains sensitive to cosine similarity cutoff selection (SGI = 20.0% at $\tau = 0.4$ versus 46.0% at $\tau = 0.7$).
6) **Geographic Concentration:** 86.8% of the LinkedIn posting corpus originates from the United States, skewing technical demand profiles toward Western, mature tech economies.
7) **Measurement Error in Behavioral Scores:** In placement benchmarks, communication scores are frequently reviewer-graded or self-reported, potentially inflating their predictive odds ratios through measurement bias.
8) **Observational Non-Causal Design:** Our econometric models establish associative rather than causal relationships; students with high project completion may possess latent motivation or unmeasured self-efficacy that drives hiring outcomes.

VI. Conclusion

This paper has presented an empirical analysis of the academic–industry skills divide using 15 datasets encompassing over 3.18 million records spanning 2020 through 2026. By integrating Natural Language Processing, Sentence-BERT semantic vector embeddings, econometric hypothesis testing, and 5-fold cross-validated predictive modeling with rigorous leakage and benchmark auditing, we have quantified the structural misalignment between university curricula and contemporary labor demands.

Our findings reveal that while surface lexical string matching exhibits a nominal 52.0% gap, dense semantic vector alignment reconciles syntax divergence, demonstrating an estimated semantic skill gap of 32.0% at baseline threshold $\tau = 0.5$ (95% CI: [23.0%, 41.0%]), with acute shortages concentrated in Data Science (18pp) and Cloud Computing (15pp). Predictive modeling demonstrates that while academic grades (CGPA) and communication proficiencies serve as vital hiring filters, applied project execution provides the decisive human capital signal associated with graduate placement. By diagnosing the structural causes of curriculum inertia and implementing agile micro-credentialing, higher education institutions and industry stakeholders can constructively align pedagogical outcomes with modern economic imperatives.

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